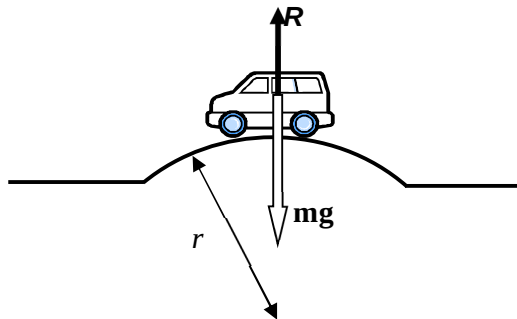


Option C is correct since the total K.E. before and after collision is the same for an elastic collision.

7
L2

Solution: B



Net force towards centre of
centripetal force

curve provides

$$mg - R = mv^2/r \quad \text{where } R \text{ is the normal contact force on car by road}$$

car loses contact with road $R = 0$

$$\Rightarrow v = \sqrt{rg} = 17.2 \text{ m s}^{-1}$$

8

Solution: B